

CS & IT ENGINEERING



Operating System

Process Management

Lecture – 1

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Recap of Previous Lecture



doubt ?

Topic

Operating System Definition

Topic

Types of Operating System



Topics to be Covered



Topic

Dual Mode of Operation

Topic

Process

Topic

Process Representation

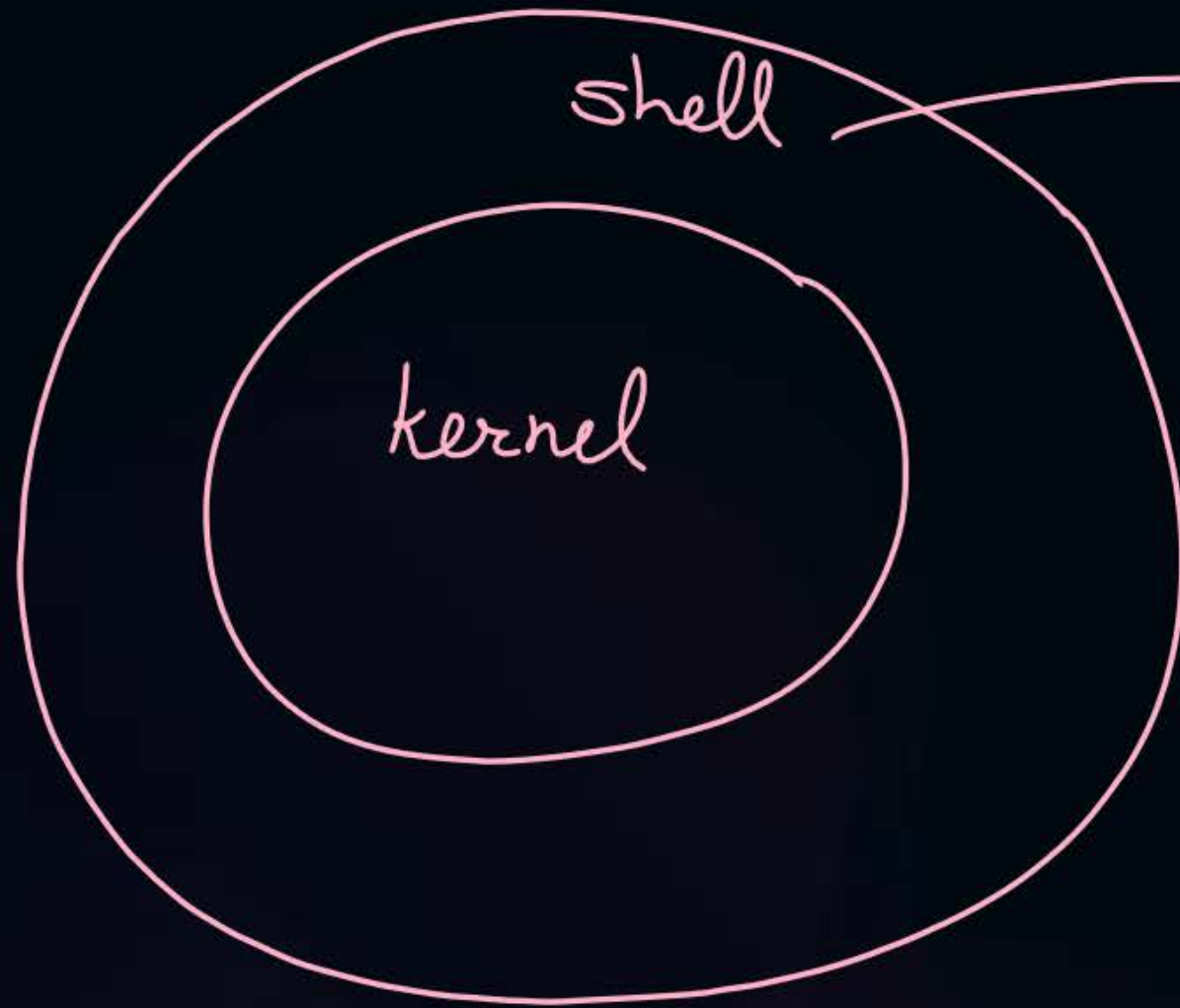
Topic

Process Control Block





Topic : Parts of OS



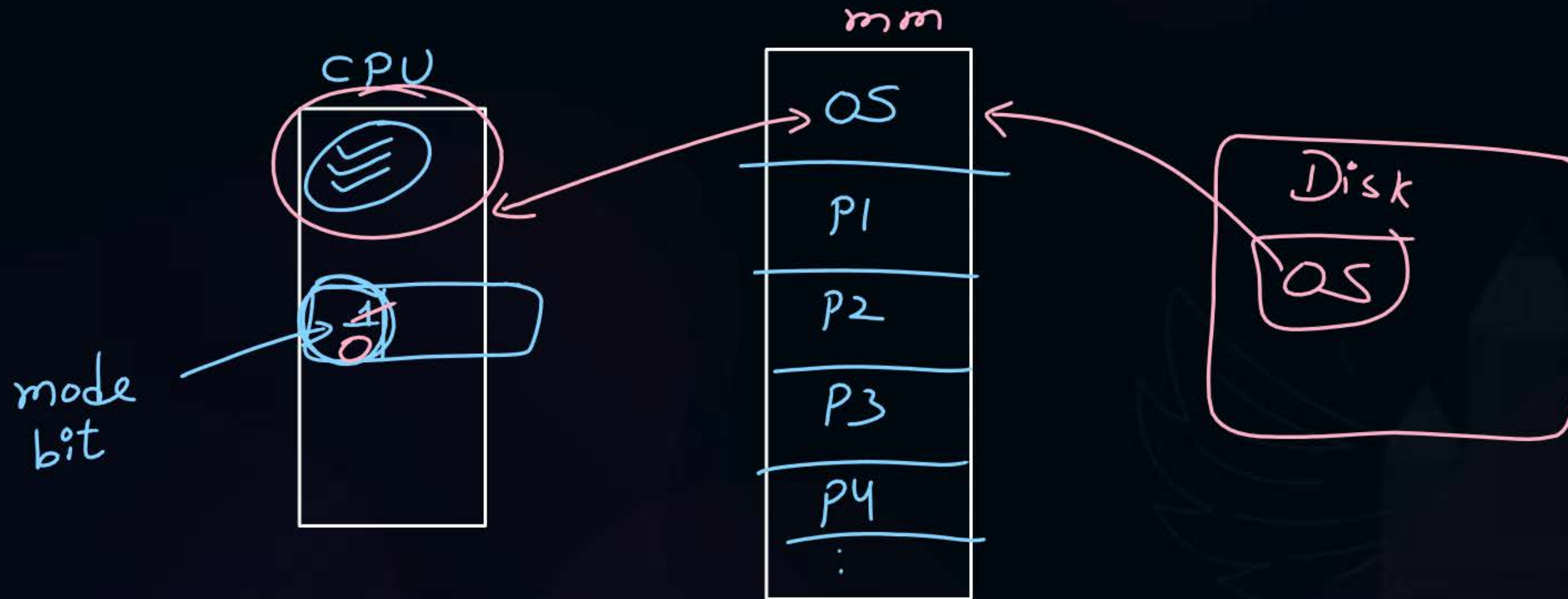
→ GUI (Graphical User Interface)
→ CLI (Command Line Interpreter)



Topic : System Call



A system call is a way for programs to interact with the operating system





Topic : Dual Mode of Operation



2 modes:

User Mode (mode bit = 1)

Kernel/System/Supervisor/Privileged Mode (mode bit = 0)

OS

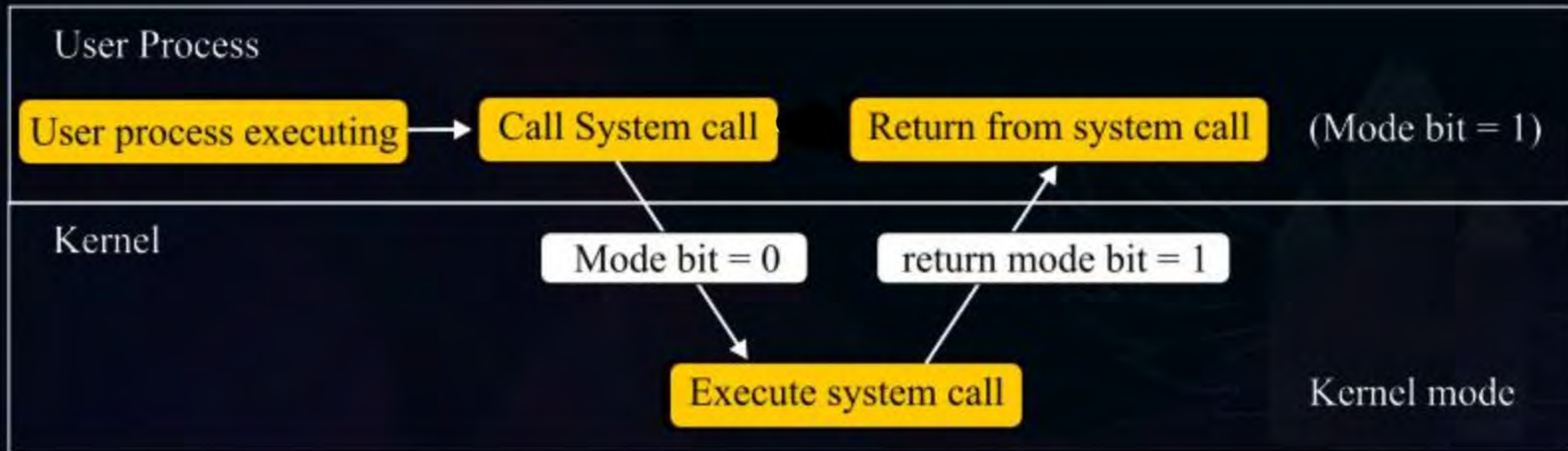


Topic : Dual Mode of Operation

↳ for system protection

2 Modes:

- User Mode (mode bit = 1)
- Kernel/System/Supervisor/Privileged Mode (mode bit = 0)





Topic : Process



Process = Program + runtime activity

flight

plane

+ source
destination

Date

dep Time
arr Time

Dur

fe

Pilot

```
void main()
{
    int a, b, c;
    a = 5;
    b = 10;
    c = a + b;
    printf("%d", c);
}
```

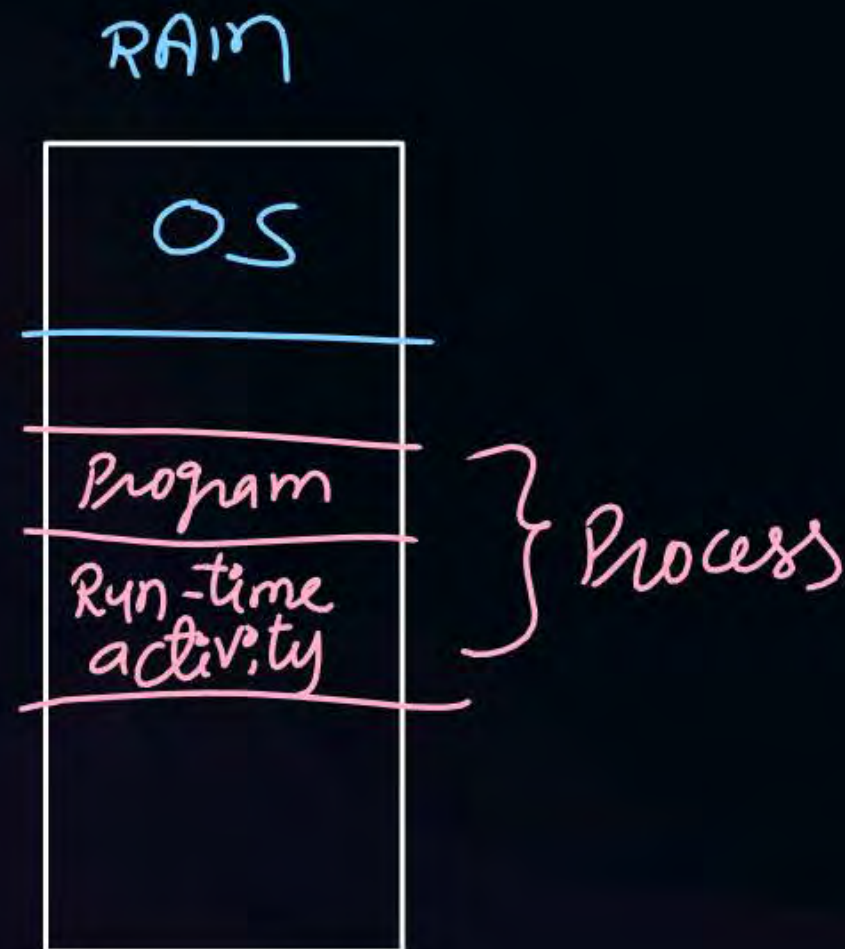



Topic : Process



□ Process:

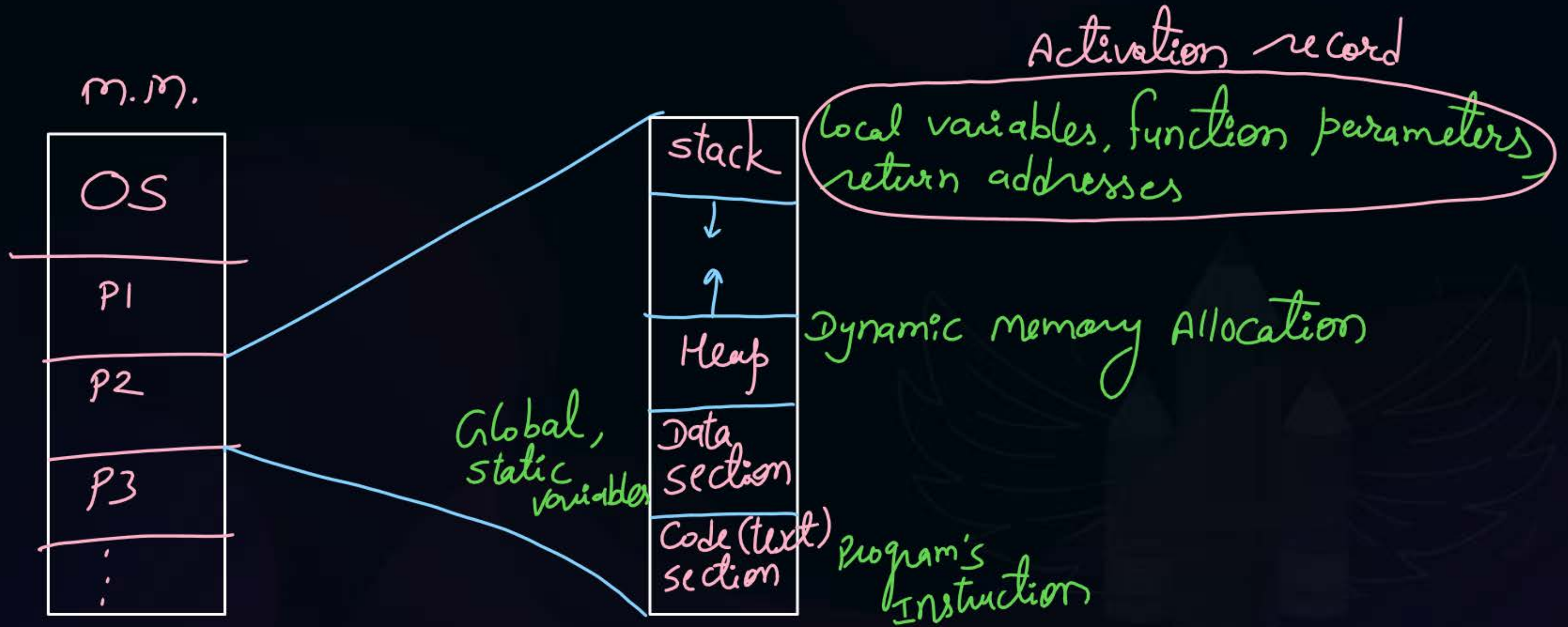
- Program under execution





Topic : Representation of a Process

each process is stored in 4 sections in m.m.





Topic : Operations on a Process



- Create (Resource Allocation)
- Schedule, Run
- Wait/Block
- Suspend, Resume
- Terminate (Resource Deallocation)





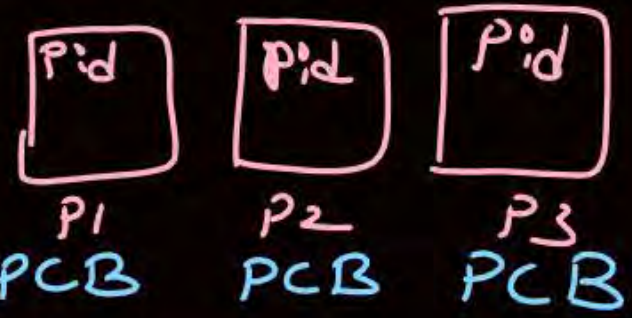
Topic : Attributes of a Process

→ os maintains attribute of each process.

- PID
- PC (Program counter)
- GPR (General Purpose Register)
- List of Devices
- Type
- Size
- Memory Limits
- Priority
- State
- List of Files

mm.

OS



p_1

p_2

p_3

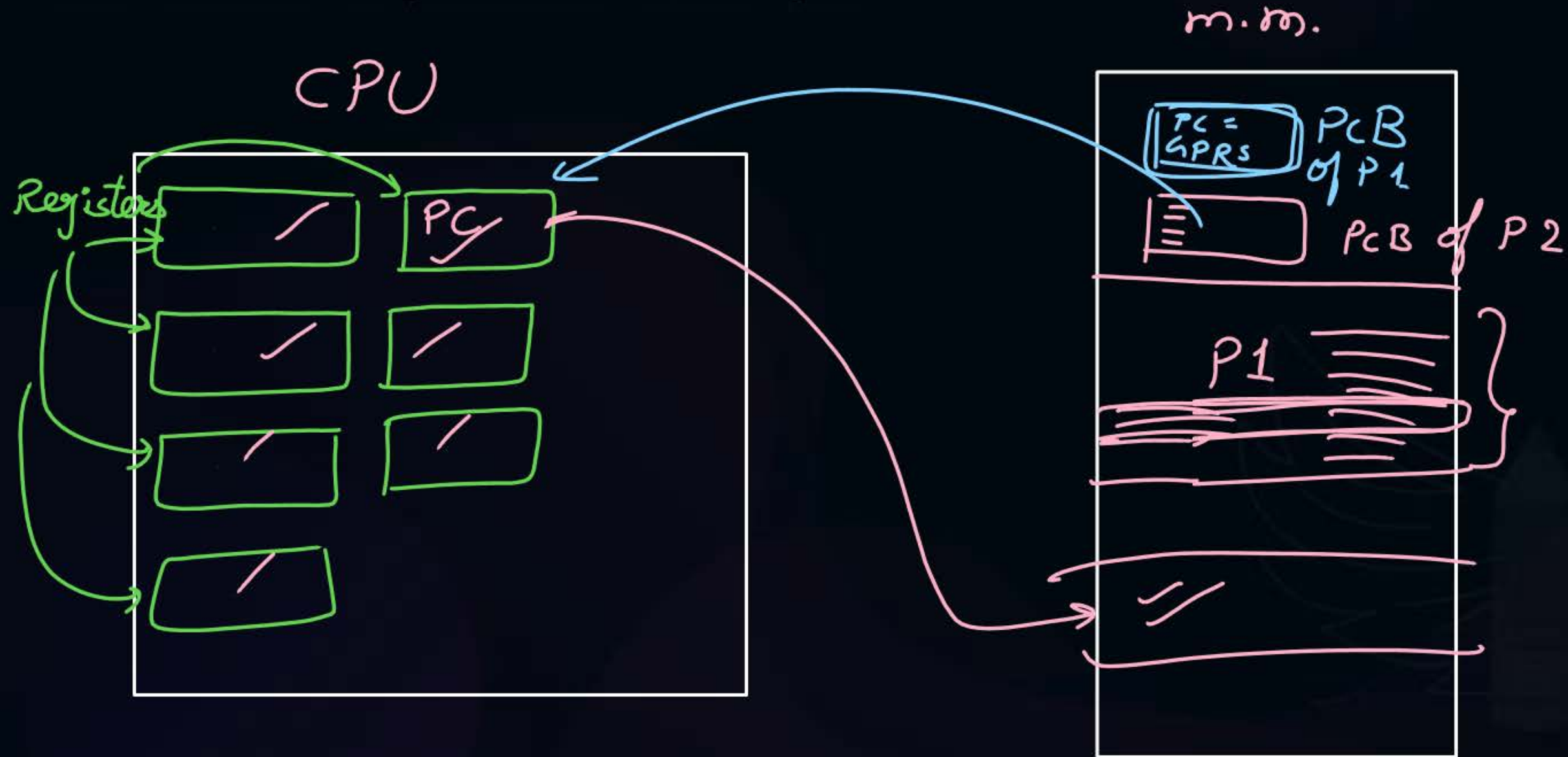
PCB $\Rightarrow$ Process
Control
Block



Topic : PCB



Also known as processor descriptor





Topic : Context



The content of PCB of a process are collectively know as 'Context' of that process

Context switch:-

storing context (Register values) from CPU to Process's PCB
for current running process.

Loading context of next process from it's PCB to CPU.

#Q. While running, a process can access its PCB from main memory?

false

[MCQ]

#Q. A process in the context of computing is:

- A** A set of instructions to be executed on a computer *→ Program*
- B** A program in execution *→ Process*
- C** A piece of hardware that executes a set of instructions *→ CPU*
- D** The main procedure of a program *→ main()*



2 mins Summary

Topic

Dual Mode of Operation

Topic

Process

Topic

Process Representation

Topic

Process Control Block

GATE-27

fundamental

→ rec.



Happy Learning

THANK - YOU

{ basics of
computer system }
→ full course

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